

Aidan Reaves

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EDUCATION

Colorado School of Mines

M.S. Computer Science

Golden, CO

2025

Colorado School of Mines

B.S. Computer Science

Golden, CO

2024

EXPERIENCE

Software Engineering Intern

Summer 2023

Pie Insurance

Denver, CO

- Built a Python-based CSV validation service to detect schema violations and data inconsistencies prior to ingestion, improving reliability and reducing manual validation effort
- Collaborated in an Agile engineering team using Git, code reviews, testing, and iterative development practices to deliver production-ready software solutions
- Worked with stakeholders to gather requirements and turn business needs into scalable technical implementations

Software Engineer (Independent)

2024 – Present

First Class Motors of Greeley

Greeley, CO

- Designing and developing a full-stack CRM system (TypeScript, React, Node.js, PostgreSQL) to replace a legacy 2007 platform, encompassing data modeling, REST API design, and database migration planning
- Drove a 50% increase in inbound leads through website management and targeted digital advertising campaigns

PROJECTS

LotManager | TypeScript, React, Node.js, Express, PostgreSQL, Docker

Jan. 2026 – Present

- Architected and built a full-stack dealership CRM supporting inventory management, customer tracking, financing workflows, and PDF document generation to replace a legacy system and modernize dealership operations
- Designed a scalable layered backend architecture (controllers, services, repositories) with 40+ REST API endpoints, JWT authentication, role-based access control, and multi-tenant dealership isolation
- Modeled complex relational business domains across 15+ PostgreSQL tables with transactions, constraints & indexing
- Built CI/CD pipelines with GitHub Actions and Dockerized deployment infrastructure on Ubuntu, including automated testing, health monitoring, and secure remote access via Tailscale
- Integrated a self-hosted LLaMA 3.2 model through Ollama to power natural language vehicle search with AI, converting plain-English queries into validated parameterized database filters to improve user experience

DreamVault | TypeScript, React, Node.js, MySQL, AWS

Oct. 2025 - Jan. 2026

- Developed and deployed a full-stack SaaS app on AWS using React, Express, MySQL, S3/CloudFront, and EB
- Implemented encrypted data storage using AES-256-GCM and searchable encrypted indexing with HMAC tokenization for secure full-text search functionality
- Built secure authentication flows with JWTs, email verification, bcrypt password hashing, protected routes, and secure cookie-based sessions
- Designed scalable REST APIs, modular backend architecture, and responsive React interfaces with real-time filtering and optimistic UI updates

Paint By Numbers Generator | Python, FastAPI, React, OpenCV, NumPy

Aug. 2025 - Oct. 2025

- Built a full-stack computer vision application that converts uploaded photos into procedurally generated paint-by-number canvases
- Implemented custom image processing algorithms including K-Means clustering, Canny edge detection, connected-component analysis, and automated palette generation
- Optimized backend image generation performance by over 80% through algorithm tuning and matrix operation improvements
- Deployed the frontend and backend using Vercel and Render with secure API communication and responsive client-side workflows

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C++, C#, PostgreSQL, MySQL

Frameworks/Tools: React, Node.js, Express, FastAPI, Docker, GitHub Actions, AWS, Tailwind CSS, Jest, OpenCV

Concepts: Full-Stack Development, REST APIs, Distributed Systems, CI/CD, Authentication & Authorization, Cloud Infrastructure, Database Design, Computer Vision